

nificant benefit of the Smart Grid 2.0 is asset management. Because of this network infrastructure approach to the smart grid, utilities can see all equipment and how the power lines interconnect that equipment to monitor the health of the systems in real time.

ISPs going out of business. If the broadband-over-power-line goals of smart grid taskforces work out as planned, one day you would just have to connect your computer to a power plug in order to surf the Web! That means, the Internet would reach every nook and corner of the country.

Energy from renewable energy sources. One of the key issues with distributing renewable energy so far has been that such sources sometimes produce less energy and even that is often seasonal, hence not justifying extensive investments in grids. However, smart grids enable such sources to be easily integrated into a flexible transmission and distribution system. Therefore we would really be able to tap the benefits of alternative energy sources, even if it is just a collection of solar panels on a company's rooftop.

Smart grids that use the cellular network. The power sector in India is held and managed by the government, but in other countries it is not so. One of the main benefits of a smart grid infrastructure is that it enables even small utility providers to prosper—it makes power a free market. Such a market with players large and small is likely to see many low-cost innovations. One such has been demonstrated by a pilot project in Texas, where a small utility has used the cell phone as the mode through which smart meters communicate with the grid. Radio frequency (RF) meshes have so far been used for this purpose in most pilot projects and implementations. The success of this innovative pilot means that the mobile network might challenge this established technology in the near future.

Load-shaping demand-response systems. Schneider, the energy stalwart, predicts that we will soon move out of the current mode of demand-response (DR 1.0) to a more sophisticated one (DR 2.0).

Current systems can manage load shedding or load curtailment (shutting off a device during peak events) and load shifting (a more sophisticated technique that moves loads away from the peaks, by preheating or pre-cooling, delaying activities such as pool pumps, defrost cycles and dishwashers).

But, DR 2.0 can do load shaping, which constantly fine-tunes demand in real time to adjust to fluctuations, such as those caused by intermittent renewables. The company feels that DR 2.0 is likely to be highly in demand in the next few years as intermittent renewables become an ever-increasing percentage of total power.

Virtual power grids. There is another highly ambitious and futuristic project from Schneider. A very simple problem led to this innovative idea. Usually, tenants, not

the owner, have to bear the electricity bills and hence the latter shows no interest in implementing any power-saving mechanism in the building. To overcome this, Schneider proposes to use its expertise to wring out significant efficiencies from large buildings. It will even bear the cost of doing so. Next, the company will offer that reduced demand to power marketers and aggregators. Of the money they make, some will be passed on to the building owners.

Schneider hopes to make money from multiple revenue streams, not just the load curtailment incentives that are part of most smart grid demand-response programmes. It also wants to tap into payments for permanent capacity (permanent efficiency improvements rather than temporary reductions during peak events); programmes for white certificate trading; and even markets for carbon reduction should they materialise in the future. In a commentary by Jesse Berst in smartgridnews.com, it is mentioned that Schneider was able to save 1½ cents per kilowatt hour when it applied the new programme to Rockefeller Center, and plans to apply the same scheme to downtown Chicago's 200 biggest buildings. The firm thinks it can pull 150 MW out of the current demand of 800 MW.

Low-cost smart meters. The cost of smart meters has been a major hurdle in the way of smart grid implementations in developing countries. However, companies like Freescale Semiconductor and Accent have demonstrated advanced system-on-chip products made especially for smart meters. This would greatly reduce the cost of smart meters.

Smart meters usually have three components: the wide-area-network connection/radio that takes data back to the utility, the home-area-network connection/radio that connects the smart home, and the metrology component that measures power, voltage, current, etc. Reducing the number of processors, by combining several functions into one, will greatly reduce the cost.

Smart grid optimisation. Optimisation is about making the smart grid work best. Some think that such optimisation is more important than smart meters, demand-response management and so on, as it improves the overall functioning of the smart grid. With multi-core processors and greater number-crunching power, intelligent smart meters that can share lots of relevant information with the utility and high-speed communication technologies, optimisation technology is facing a boom time. Several companies such as GRIDiant, Lockheed Martin, Ambient, S&C Electric and even Oracle have major plans in the optimisation space. With jazzy dashboards and powerful energy management solutions, these optimisation suites are targeting a 'power'ful market! Over time, these tools can help to completely automate the entire monitoring, control and management tasks in a smart grid. ●

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